SRW3 Phase 1L — Universal Induction Mechanization Report

The induction step mechanized where the backend allows; the universal theorem honestly NOT MECHANIZED, with the exact blockers
Branch `phase1l-universal-induction` (from the corrected Phase 1K commit `b1dbe90`, tag `phase1k-r1-complete`) · additive tree `phase1l/` · frozen base `phase1k-implementation-package` @ `b1dbe90` (frozen 1J semantics UNCHANGED)

1. Objective and outcome

Phase 1L's mandate (the handoff, PHASE1K_AUDIT_FINDINGS.md resolved first; SRW3_Phase1L_Universal_Induction_Handoff.md): mechanize the Phase 1J reachable-committed-state safety theorem for the already-declared finite policy fragment and the already-declared four-operation `EVM_F` fragment — with the explicit rule that the phase may be declared complete ONLY if the universal theorem is actually mechanized, and that a blocked Haskell proof must be RETAINED, its exact blocker identified, and the strongest truly proved theorems reported.

Outcome (Level III, exact): the universal all-path theorem (`J-REACHABLE-COMMITTED-SAFE` class) is **NOT MECHANIZED**. The ladder mechanizes the induction STRUCTURE around it: 14 claims PROVED over the frozen J machine (the anchor/config/catalog frames, the pre-init family, the refusal family, the no-commit content of the source-refusal class on every resolution path, and the four commit-rejection/freshness theorems), 16 attempted claims retained with their raw failures and the blockers named to the kore residual level, one FROZEN-SEMANTICS FINDING reported with the frozen 1J T11 transcript as its counterexample witness, all negative controls failing exactly as designed, and the concrete substitutes (Python prefix enumeration, the fresh LLVM T-series byte-identical to the frozen records) re-verified from the corrected base. No result is relabeled; no consensus-integration or full-EVM claim is made.

2. What was built (additive `phase1l/` only)

```
phase1l/
semantics/j_invariant.k the invariant predicate (JInvJ /
JInvPreinitJ / JChainCoherentJ / JInvPlusJ)
+ the PROOF-FACING total wrappers (§5 — the
explicit definedness assumption); imports the
FROZEN J machine; NO semantic change
proofs/l_proofs.k the 30-claim ladder (§0 experiments, §A
frames, §B pre-init, §C step lemmas,
§C2 dispatch resolution, §D commit family,
§E the universal attempts)
proofs/run_l_proofs.sh the per-claim runner (expected-outcome
gating; negative controls MUST fail)
transcripts/k/ 30 kprove logs + claims-status.csv +
claims-classification.csv + the fresh 1J
re-run record + README
transcripts/llvm/ the FULL frozen 1J T-series re-run fresh
(T0-T13 + 5 kparse rejections + anchor) —
byte-identical to the frozen records modulo
header lines
transcripts/attacks/ run_l_attacks.txt (56/56 PASS) + the fresh
1J unittests (27/27) + attack matrix (48/48)
audit/ induction-diagnosis.md (the blocker census),
rule-inventory.md, findings.md
(L-FINDING-1), assumption-register.md
python/run_l_attacks.py the L-A01..L-A20 tier over the FROZEN mirror
provenance/ input hashes + reproduction commands
report/ this file (MD + PDF)
```

The reachable rule set of the 1L proof definition is byte-for-byte the frozen J machine's (`audit/rule-inventory.md` cross-checks every rule; the 1L module adds `[function, total]` definitions only).

3. The K/LLVM toolchain and the fresh regression record

- Toolchain rebuilt from the pinned recipe of record after the sandbox reset: K v7.1.337 (jammy deb), z3 4.13.3, flex 2.6.4, clang/LLVM-15 15.0.6-4+b1 (+ `libclang-common-15-dev` extraction, `lld-15`, `libmpfr-dev`/`libsecp256k1-dev` headers), pristine `llvm-kompile-clang` driver with the single path substitution (the R2 patch NOT applied). Environment repairs recorded: a java wrapper capping the frozen build scripts' `-Xmx6g` (this 4-GiB container OOMs); `LIBRARY_PATH` defaulted in env.sh (`set -u`).
- **Fresh package verification from the corrected 1K base** (the sequencing precondition): `install.sh --adopt-quarantine` exit 0; `run_all.sh` exit 0 (unittests 27/27; 1J attacks 48/48 incl. the reachability enumeration 3474/386/0; R1 48/48, R2 87/87, R3 181/181 with K↔Python cross-checks; 1E/1F/1G green); the vector semantic-equivalence check PASS (23/23 fields); `verify_all.sh` PASS post-restoration (55/55) — the Phase 1K-R1 corrective evidence.
- **The frozen 1J kprove ladder re-run fresh** from the corrected base: 38/38 claims that PROVED at 1J reproduced PROVED (the 4 retained-class claims re-run bounded by the wrapper budget; their frozen full-run logs stand as the record — the 1K disclosure carried forward).

- **The frozen 1J LLVM T-series re-run fresh** from the package sources: 14 executed demos + 5 kparse rejections, and the decisive T0 certificate, T2 honest two-block pipeline, T6 unsafe-composition rejection, and the self-computed anchor are BYTE-IDENTICAL to the frozen 1J records modulo the header lines.
- **The 1J Python suites re-run fresh**: unittests 27/27 OK; the attack matrix 48/48 PASS.

4. The K proof ladder (the raw discipline)

30 claims, one definition, one canonical pass, per-claim exit codes and retained logs (`transcripts/k/`, `claims-status.csv` + the scientific classification in `claims-classification.csv`):

- **PROVED (14)** — the frames (anchor/config/catalog untouched by the dispatch step), the pre-init family (nothing pinned, no receipts, no committed history; refused commands change nothing), the refusal family (unsupported/bounds write only the diagnostic and mint nothing; the source-refusal class moves no committed cell and mints no receipt on EVERY resolution path — including the stuck path), and the commit-rejection family (stale slot, consumed receipts, forged verdicts, prior-slot exec receipts — the freshness ties decided CONCRETELY).
- **NOT MECHANIZED (11, blocked — attempts retained)** — the invariant-carrying steps that must materialize hash-rooted builders over symbolic arguments: the gate/exec mint steps, the accept-poststate step, the symbolic-config init base, and the two universal compositions (all-path, committed-prefix).
- **FINDING WITNESS (1)** — L-INV-EXEC-DISPATCH-SOURCE (see §6).
- **NEGATIVE CONTROLS / EXPERIMENTS (4, all failed as designed)** — the destination-constraint idiom pair (B2), the SafeJ-removal control, and the weakened-invariant all-path control (the pair that exposes a vacuous circular closure).

Score: 14 PROVED / 0 gate alarms over the full inventory; the runner exits 0 when every raw outcome matches its expectation — the retained failures are the DESIGNED honest outcomes, not silent skips.

5. Why the universal theorem is NOT MECHANIZED (the blocker census)

`audit/induction-diagnosis.md` carries the full census; the four blockers:

- **B1 — the hash-definedness boundary (dominant).** kore discharges `#Ceil(Keccak256raw(x))` only over fully-literal arguments. Every blocked claim's final residual is an undischargeable Ceil over a symbolic (or hash-containing) argument — probed minimally and confirmed by the ladder's own residuals. Total wrappers and reflexivity hypotheses do NOT rescue it. Every hash EQUALITY remains a claim hypothesis where it is one; the boundary is about materializing hash applications, not about any assumed hash property.
- **B2 — the destination-constraint idiom is unusable** (`requires` is evaluated at the source; destination-only variables make it stuck) — the experiment pair documents it; the ladder therefore carries the invariant by shared cells and exact destination terms.
- **B3 — the pool-Ceil class** — set well-formedness over symbolic receipt builders; concrete pool shapes (`.Set`) refute the accept unification and prove — the working convention applied systematically.
- **B4 — the stuck-gate path** — a pGateEvalJ with unmet mint preconditions matches NO rule (documented non-commit, handoff §3 item 5); a stuck configuration does not match the all-path claim's source pattern, so the circular re-application dies.

The [circularity] forall-n recipe (the Phase 1B GH-T2 precedent) does not transfer: 1B's invariant was decidable arithmetic on concrete cells, so the circular re-application VERIFIED it; the J machine's `SafeJ` over symbolic maps is a stuck term the re-application can only ASSUME — a closure would be vacuous w.r.t. safety, exactly what the paired negative control was run to detect.

6. The finding against the frozen semantics (L-FINDING-1)

The frozen unauthorized-source dispatch rule's `<jlineage>` pattern is the single-entry map `-1 |-> cfg` with NO tail variable: in a fully-pinned machine the rule cannot match, so an unauthorized-source execution is STUCK — the `jrefusedSource()` diagnostic never fires (the frozen 1J T11 transcript shows exactly this stuck configuration). Safety is preserved (stuck = no transition = no receipt = no committed movement; PROVED as L-INV-EXEC-SOURCE-NOCOMMIT), but the documented explicit-refusal expectation is impaired and the K machine diverges from the Python mirror on this one resolution (both fail-closed). Per the handoff §3 protocol the model is NOT repaired inside Phase 1L (the 1J tree is frozen); the repair (adding the map-tail variable) belongs to a separately justified commit.

7. The theorem, stated exactly (and its boundary)

Under the assumptions of the frozen 1J surface (the authenticated external anchor; Keccak collision resistance; frozen-gate soundness; the registered finite invariant language; the declared execution-source authenticity boundary — `audit/assumption-register.md`), and for the DECLARED finite policy fragment and the four-operation `EVM_F` fragment:

- **MECHANIZED (this phase):** the invariant predicate over the actual machine configurations; the frame, pre-init, refusal, and commit-rejection/freshness step families; the freshness ties that rule out cross-commit receipt reuse; the structural rule inventory (only the accepted commit writes the state, and it writes exactly the term its guard certified).
- **NOT MECHANIZED (honest):** the universal composition — "for every finite execution beginning in the defined initial configuration with an authenticated anchor, every configuration reached after successful initialization satisfies the J safety invariant". The base and the per-class steps are mechanized to the backend's boundary; the composition itself is blocked by B1-B4. The 1J concrete substitutes (the 3,474-prefix enumeration with 386 unsafe derived states caught and 0 committed unsafe; the LLVM honest/unsafe pipelines) remain DEMONSTRATED evidence, not proof.
- **NOT CLAIMED:** full-EVM trace authenticity, arbitrary-policy closure, anchor governance, Ethereum consensus enforcement, or any statement that "Ethereum applications cannot weaken foundational security". Level III only; no consensus-integration claim.

8. Completion gate (handoff §8), item by item

gate item	disposition
1K verification fresh clean install/run/verify; mismatch resolved; verify fails on required-field omissions	the Phase 1K-R1 corrective commit `b1dbe90` (tag `phase1k-r1-complete`; `phase1k-complete` preserved at `d32ad13`); PASS
1L starts from the corrected 1K commit, additive `phase1l/` only	branch base `b1dbe90`; git diff outside `phase1l/` EMPTY (worklog update only)
frozen trees byte-identical to the 1K base	verified (`git status` clean; the transient in-place transcript regeneration for the LLVM re-run was restored byte-exact before commit — the established 1I/1J pattern)
every K claim: exit code, full log, raw result, classification	30/30 (`claims-status.csv` + `claims-classification.csv`)
the invariant not assumed in its own induction step	SafeJ enters only via the computed guards / shared cells; the negative control verifies the dependency

gate item	disposition
all transition classes accounted for	rule-inventory.md + the per-class claims (init/exec/dispatch/gate/reject; the stuck gate case = B4 disclosure + L-A19)
all-path theorem mechanically proved, or explicitly NOT MECHANIZED with a minimal retained failure and a clear explanation	**NOT MECHANIZED**; the attempts retained; the blocker census at kore-residual level
Python enumeration and LLVM demos pass unchanged; not substituting for induction	48/48 + 27/27 fresh; T0/T2/T6 + anchor byte-identical fresh; classified DEMONSTRATED only
proof reports distinguish PROVED / DEMONSTRATED / CHECKED / ASSUMED / NOT MECHANIZED	the classification CSV + this report
no consensus-integration or full-EVM claim	§7

9. Artifact index

`phase1l/semantics/j_invariant.k` · `phase1l/proofs/{l_proofs.k, run_l_proofs.sh}` · `phase1l/transcripts/k/` (30 logs + status + classification + the 1J fresh re-run record + README) · `phase1l/transcripts/llvm/` (the fresh T-series) · `phase1l/transcripts/attacks/` (L-attacks 56/56; the fresh 1J suites) · `phase1l/audit/` (induction-diagnosis, rule-inventory, findings, assumption-register) · `phase1l/python/run_l_attacks.py` · `phase1l/provenance/` · this report (MD + PDF).